

About Validation Execution Standard

Governing the Difference Between Validation Designed and

Validation Actually Performed

A validation methodology can be excellent.

The criteria can be precise.

The evidence can be fit.

The sources can be reliable.

And the validation can still fail methodologically if the required validation process was not actually performed as governed.

Validation Execution Standard (VEXS) is the seventh Parent Standard of **VALIDOS® — Validation Governance Architecture**, governing the operational transition from an approved validation design to demonstrable validation activity.

VEXS addresses one fundamental question:

Was the approved validation process actually performed as governed?

Why Validation Execution Standard Exists

Validation does not occur because a methodology exists.

It occurs when that methodology is applied to a specific Validation Object under actual operational conditions.

Between:

What should happen

and:

What actually happened

there may be substantial differences.

A validation process may:

- skip required steps,
- examine the wrong object version,
- use different evidence,
- operate outside the approved scope,
- substitute instruments,
- change environmental conditions,
- alter sequencing,
- experience interruptions,

- introduce manual interventions,
- or fail to preserve sufficient execution records.

Without execution governance, a later validation result may appear methodologically legitimate even though the process producing it materially departed from the approved validation structure.

VEXS exists to make that difference governable.

Validation Design Is Not Validation Execution

VALIDOS® deliberately separates:

Validation Method

from:

Validation Execution

Validation Method asks:

How should validation be performed?

Validation Execution asks:

What was actually performed?

This distinction is fundamental.

A method is a governed design.

Execution is its operational realization.

Therefore:

Approved Method ≠ Executed Validation

From Methodology to Operational Reality

Before VEXS, VALIDOS® has established:

Validation Object

Validation Purpose & Scope

Validation Criteria

Validation Method

Validation Evidence Fitness

Validation Source Reliability

Together, these define the governed validation basis.

VEXS moves that structure into operational reality.

The lifecycle becomes: **Validation Design** → **Execution Basis** →
Execution Authorization → **Execution Instance** → **Actual Validation**
Activity → **Execution Record**

Only after this transition can VALIDOS® legitimately ask what the validation demonstrated.

The Correct Object Must Be Executed Against

Validation must remain attached to the object it was designed to examine.

If validation was approved for:

Object Version 3.2

but execution occurs against:

Object Version 3.4

the difference may be material.

Likewise, changes in:

- configuration,
- system state,
- dependencies,
- environment,
- operating mode,
- or object boundaries

may change what was actually validated.

VEXS therefore preserves Validation Object continuity throughout execution.

Scope Must Survive Execution

A Validation Scope defined during planning can change during actual execution.

Some areas may become unavailable.

Tests may be omitted.

Additional areas may be examined.

Conditions may force narrower coverage.

VEXS makes these changes explicit.

A later Validation Determination should never claim a scope broader than the validation activity actually performed.

Criteria Must Become Activities

Validation Criteria are not operationally meaningful merely because they appear in a validation document.

They must be connected to actual validation activities.

VEXS therefore asks:

Which criteria were actually evaluated?

Through which activities?

Using which evidence?

Under which conditions?

Were any criteria omitted?

VEXS does not yet decide whether those criteria were satisfied.

It establishes that the required evaluation actually occurred.

Planned Execution and Actual Execution

VALIDOS® creates an explicit distinction between:

Planned Execution

and:

Actual Execution

This allows the validation process to compare:

Expected Activity → Actual Activity

rather than assuming they are identical.

Where material differences exist, they become governed Execution Deviations.

Execution Conditions Matter

Validation does not occur in abstraction.

It occurs under conditions.

These may include:

- physical environment,
- digital environment,
- system configuration,
- network conditions,
- operational load,
- population,
- geographic context,
- human involvement,
- available tools,
- simulation settings,
- or system state.

If those conditions materially differ from those required by the Validation Method, the difference may affect execution legitimacy.

VEXS therefore makes execution conditions part of the validation record.

Sequence Can Matter

Some validation procedures depend upon order.

A test performed after a system has already been modified may not establish the same thing as the same test performed before modification.

Likewise, one validation activity may depend upon another being completed first.

VEXS therefore governs sequence where sequence has methodological significance.

Evidence Must Survive the Transition Into Execution

Evidence that was previously qualified by VALIDOS® should not lose its governance conditions once execution begins.

VEXS preserves relationships to:

- Evidence Fitness,
- evidence version,
- evidence limitations,
- evidence restrictions,
- Source Reliability,
- source conditions,
- and applicable corroboration requirements.

This prevents execution from silently substituting governed evidence

with materially different evidence.

Reliable Sources Can Still Be Used Incorrectly

Source Reliability does not guarantee correct execution.

A reliable source may be:

- used outside its approved role,
- used beyond its reliability boundary,
- interpreted incorrectly,
- substituted,
- or used without required corroboration.

Therefore:

Reliable Source ≠ Correct Execution

VEXS governs how source conditions are actually respected during validation activity.

Execution Can Be Human, Automated, or Hybrid

Modern validation may involve people, software, AI systems, autonomous agents, instruments, or combinations of them.

VEXS is execution-model neutral.

Human Execution

A person may:

- perform tests,
- make observations,
- interpret evidence,
- operate instruments,
- review outputs,
- or intervene in the validation process.

Automated Execution

A system may:

- execute tests,
- monitor behavior,
- process evidence,
- run simulations,
- generate measurements,
- or evaluate defined conditions.

Hybrid Execution

Humans and machines may divide responsibility across the same validation process.

VEXS governs the actual execution structure regardless of who or what performs it.

Execution Deviations Are Governance Events

Real-world validation rarely proceeds exactly as planned.

Deviation itself does not automatically invalidate validation.

The important questions are:

What changed?

Why did it change?

Was it authorized?

What did it affect?

Was the effect material?

Were compensating controls required?

VEXS transforms deviation from an undocumented irregularity into a governed validation event.

Not Every Deviation Has Equal Importance

VALIDOS® may distinguish:

Immaterial Deviation

Minor Deviation

Material Deviation

Critical Deviation

Execution-Invalidating Deviation

This prevents two opposite errors:

treating every deviation as catastrophic,

and:

ignoring deviations that materially undermine validation.

Authorization Does Not Erase Methodological Effect

A deviation may be formally authorized and still materially affect validation.

Therefore: **Authorized Deviation ≠ Methodologically Harmless Deviation**

Authorization establishes that the change was governed.

Its methodological consequences must still remain visible.

Substitution Must Be Visible

Real validation processes may substitute:

- evidence,
- sources,
- instruments,
- datasets,
- models,
- personnel,
- environments,
- procedures,
- or controls.

Substitution may be entirely legitimate.

But equivalence should not be assumed merely because the substitute appears similar.

VEXS requires material substitutions to remain identifiable.

Interruptions Matter

Validation may be interrupted by:

- system failure,
- infrastructure outage,
- missing evidence,
- safety conditions,
- human intervention,
- environmental change,
- resource failure,
- or unexpected system behavior.

The governance question becomes:

Can execution continue?

Must part of it be repeated?

Must it restart entirely?

Can the interruption be accepted conditionally?

VEXS provides the execution layer in which those questions can be answered.

Failure to Validate Is Not Validation Failure

This is one of the most important distinctions in VEXS.

Suppose a required test cannot be completed because the testing environment fails.

That does not demonstrate that the Validation Object failed the criterion.

It demonstrates that validation could not establish the required result.

VALIDOS® therefore establishes: **Failure to Validate ≠ Validation Failure of the Object**

This prevents methodological failure from being incorrectly transformed into an object-level conclusion.

Execution Completeness

VEXS asks whether all materially required validation activities were actually performed.

Execution may therefore be:

Complete

Conditionally Complete

Partially Complete

Incomplete

or:

Completeness Undetermined

Completeness concerns the validation process.

It does not indicate whether the object passed or failed.

Execution Conformity

VEXS separately asks whether actual execution sufficiently corresponded to the approved validation basis.

Possible conditions may include:

Execution Conformant

Execution Conditionally Conformant

Execution Conformity Undetermined

Execution Non-Conformant

Execution Invalid

This creates a formal governance gate before Validation Determination.

A Perfectly Executed Validation Can Produce a Negative Result

This distinction is essential.

A validation process may be:

- complete,
- perfectly conformant,
- fully documented,
- and methodologically strong

and demonstrate that the Validation Object fails.

That is not failed validation execution.

It is successful execution producing a negative validation finding.

Therefore:

Execution Conformant ≠ Validation Object Valid

An Apparently Valid Object Does Not Repair Invalid Execution

The opposite is equally important.

An object may in reality satisfy all required conditions.

But if the validation process used to establish that conclusion was materially invalid, VALIDOS® should not manufacture methodological confidence from the likely correctness of the result.

Therefore:

Execution Must Be Reconstructable

A strong validation process should allow a later reviewer to reconstruct materially:

What was validated?

Which version?

Against which criteria?

Using which method?

Using which evidence and sources?

Who or what performed the activities?

Under which conditions?

In what sequence?

What deviations occurred?

What outputs were generated?

This reconstruction creates the bridge between validation activity and auditability.

Execution Records Become Methodological Evidence

The execution process itself creates evidence.

This may include:

- logs,
- observations,
- test outputs,
- measurements,
- timestamps,
- system records,
- operator records,
- simulation outputs,
- deviation records,
- interruption records,
- and other execution artifacts.

Together these form the **Validation Execution Record**.

The record establishes what the validation process can actually

demonstrate was performed.

Execution Can Be Repeated

A failed or materially deficient execution does not necessarily end validation.

VEXS allows governance of:

Re-Execution

and where methodologically defensible:

Partial Re-Execution

This permits affected validation activities to be repeated without automatically discarding unaffected valid execution.

Historical Execution Must Remain Visible

Repeated execution should not overwrite earlier validation history.

Where validation is repeated after:

- failure,
- deviation,
- system change,
- evidence change,
- or corrective action,

the relationship between executions should remain traceable.

This creates validation execution continuity.

Determination Readiness

VEXS ultimately asks whether execution has reached a state where a legitimate Validation Determination can begin.

This requires sufficient understanding of:

- object continuity,
 - scope conformity,
 - criteria execution,
 - method conformity,
 - evidence continuity,
 - source conditions,
 - deviations,
 - completeness,
 - conformity,
 - and execution traceability.
-

Where those conditions are sufficiently established, execution can progress.

Where they are not, VEXS can prevent premature determination.

Determination Block

A critical execution deficiency may create a **Determination Block**.

This means VALIDOS® does not permit methodological uncertainty about execution to disappear merely because an organization wants a final answer.

Possible blocking conditions include:

- wrong Validation Object,
- critical missing activity,
- invalid method substitution,
- critical scope deviation,
- irrecoverable evidence discontinuity,
- critical execution failure,
- or insufficient execution traceability.

The validation process must first resolve the execution problem or explicitly preserve the inability to determine.

Position Within VALIDOS®

Validation Execution Standard is the seventh Parent Standard of **VALIDOS® — Validation Governance Architecture**.

Its architectural position is:

Validation Object → Validation Purpose & Scope → Validation Criteria → Validation Method → Validation Evidence Fitness → Validation Source Reliability → Validation Execution → Validation Determination → Validation State → Validation Reliance & Revalidation

The first six Parent Standards establish the governed basis upon which validation can operate.

VEXS establishes whether that basis was actually transformed into a legitimate validation execution.

Only then does VALIDOS® move to the next question:

What does the governed validation record actually demonstrate about the Validation Object?

Relationship to Other OOF® Architectures

VEXS operates alongside complementary OOF® architectures.

GOA™ may establish authority, delegation, and governance responsibilities affecting validation execution.

OBIDENTITY® may preserve identity, origin, ownership, provenance, and continuity relationships.

INTEGROS® may provide integrity conditions affecting validation activities and execution records.

ORA™ may establish complementary operational-reality conditions.

AGA™ may establish accountability for execution activities, deviations, interventions, and decisions.

AIG®, CLIA®, MGIA™, and **ASGA™** may provide specialized governance conditions where AI behavior, cognition, memory, or autonomous execution materially affects validation.

RIS™ may inform the depth, strength, and control requirements appropriate to the consequence of validation failure.

VEXS does not duplicate these architectures.

It consumes relevant governed information and answers one validation-specific question:

Was the validation process actually executed with sufficient conformity, completeness, control, continuity, and traceability to support a legitimate Validation Determination?

Why It Matters

Validation is often documented more strongly than it is executed.

Policies describe ideal processes.

Methodologies define required procedures.

Testing plans specify controlled conditions.

Standards define expectations.

But operational reality may differ.

VALIDOS® therefore refuses to infer execution from documentation.

It requires the distinction:

Designed Validation

versus:

Executed Validation

to remain visible.

From Validation Procedure to Validation Proof

Traditional validation may ask:

Was the procedure defined?

VEXS adds:

Was it actually performed?

Against the correct object?

Within the correct scope?

Under the required conditions?

Using the governed evidence?

With the required sequence?

What changed during execution?

Can we reconstruct what happened?

Is the execution sufficiently conformant to support determination?

This transforms execution from an assumed procedural step into an independently governable validation layer.

Foundational Principle

A validation method has no operational validity until its required validation activities are actually performed.

Canonical Closing Statement

Validation Execution Standard establishes the operational performance layer of VALIDOS® — Validation Governance Architecture by separating validation design from validation actually performed. Through governance of execution authorization, instantiation, object continuity, scope conformity, criteria execution, method conformity, evidence and source continuity, conditions, participants, sequencing, deviations, substitutions, interruptions, failures, completeness, conformity, reconstruction, repetition, and

traceability, VEXS ensures that Validation Determination rests upon demonstrable execution rather than methodological intention, procedural assumption, or undocumented operational activity.

Parent Resources

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Related Documents

[→ Validation Execution Standard](#)

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

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